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CS2024 - Theory of Computation

nonRegular Expressions

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1.

A nonregular Language

There is nothing we can do

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Previously...

- We can build, ANY language with a **Finite** Automaton...
  - ***... or can we?***
- Can we build an DFA of $L = \{0^n1^n, n \geq 0\}$?
  - Or a NFA?
  - Or a Regular expression?
- ***We*** *can't...*

Previously...

- ***We*** *can't...*
  - But just because we can't doesn't mean NONE of the infinite DFA recognizes L
  - We need to **convince** this particular language is not regular.
  - HOW?
  - By showing there is an invariant among all regular languages, that just so happens L doesn't have.
- This is where pumping Lemma appears

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2.

The Pumping Lemma

It's not pompeo e.e

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Pumping Lemma

Let A a regular language, then there is a number p (pumping value) where

- For any string $s \in A$, with length at least p, s can be divided into 3 pieces $s = xyz$, such that
  - $|y| > 0$
  - $|xy| \leq p$
  - $xy^iz \in A$, for all $i \geq 0$

HOW we will use PL?

- In spanish, pumping lemma tells us that all regular language A has the property that
  - a string large enough, can be “pumped” and generate infinitely more strings that also belong to A
- If some language doesn’t have that property, we can use proof of contradiction to show that language is not regular

Proof , $L = \{0^n 1^n, n \geq 0\}$

Assume by contradiction that L is regular

- This means we can apply pumping lemma to L
- Exists a pumping value p, such that any string s $s \in A$, with length at least p, s can be divided into 3 pieces s = xyz, such that $|y| > 0$, $|xy| \leq p$ and $xy^iz \in A$, for all $i \geq 0$
- in particular, the word $s = 0^p 1^p$ also can be divided like that
- Since $|xy| \leq p$, the block **y** only contains 0s, and since $|y|>0$, it has at least one 0.
- This means xyyz has more 0s than 1s, and can't be on A
- We reach a contradiction

This proves L is not regular

Exercise

Proof the following languages are not regular. $\Sigma = \{0,1\}$

- $\{w: w \text{ has the same number of 0s and 1s}\}$
- $\{uu: u \in \Sigma^*\}$
- $\{1^k, \text{ where } k \text{ is a square number}\}$
- $\{0^i 1^j: \text{ with } i>j\}$
- $\{1^k, \text{ where } k \text{ is a prime number}\}$

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Thanks

Let’s keep finding mistakes

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